

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S7 (R, S) / S7 (PT) (R) Examination December 2023 (2019 scheme)

Course Code: MCN401**Course Name: INDUSTRIAL SAFETY ENGINEERING****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

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| 1 | With suitable examples differentiate between unsafe act and unsafe conditions referred to construction industry. | (3) |
| 2 | How do voluntary agencies supports safety measures in various organizations? | (3) |
| 3 | With suitable sketches mention the features of any two PPE for eye protection in industrial hazardous sites. | (3) |
| 4 | Explain the major objectives of work permit systems. | (3) |
| 5 | Explain any five critical reasons for high rate of accidents in construction industry. | (3) |
| 6 | As an engineering supervisor how do you ensure the overall safety during demolition operations? | (3) |
| 7 | Explain the key features of hearing conservation program in production industries. | (3) |
| 8 | Explain the safety guidelines to be followed for the lifting hooks used in heavy industrial mechanisms. | (3) |
| 9 | Explain the different causes of electric hazards. | (3) |
| 10 | Write short note on toxic gas release due to chemical and fire explosion. | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

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| 11 | a) Explain Heinrich's domino theory for accident causation. Also mention any two modifications made on this basic concept. | (7) |
| | b) Explain the six distinguished functions of safety committee in an industry. | (7) |

OR

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| 12 | a) With suitable examples explain how does safety management improve productivity of a firm. | (6) |
| | b) Explain the responsibilities of supervisors to ensure safety in an industrial organization. | (8) |

Module II

- 13 a) Explain the key functions of any three different types of respiratory protective equipments. (6)
- b) Explain any ten important responsibilities of housekeeping head of an organization. (8)

OR

- 14 Write short notes on the following: (14)
- i. Cold work permit
 - ii. Reactive measures for safety performance monitoring
 - iii. Five S (5S) concept in housekeeping.

Module III

- 15 Write short notes on the following: (14)
- i. Cumulative Trauma Disorders due to ergonomic issues in work sites
 - ii. Hazards and risks during Underpinning works
 - iii. Safety measures during excavation works

OR

- 16 a) With suitable schematics, mention any eight specific safety precautions for scaffolding to ensure the safety of the workers. (9)
- b) Explain the classification of residential buildings based on the National Building Code of India (5)

Module IV

- 17 a) Describe about the safety practices to be followed during arc welding process. (8)
- b) Differentiate between manual and mechanised material handling with suitable examples. (6)

OR

- 18 a) An industrial plant is equipped with crane, conveyors, hoist etc. Explain the maintenance strategy to be followed for any of the material handling equipment to ensure safe working environment. (9)
- b) With suitable schematics, explain the operation of any presence sensing device suitable as a safety guarding device for an industrial unit. (5)

Module V

- 19 a) Describe the relevance and procedure for HAZOP for safety assessment. (9)
- b) Explain how does MSDS differ from a product label referred to material safety. (5)

OR

- 20 a) Describe about the classification of fires. Also explain any four types of fire extinguishers. (9)
- b) Describe about any four hazardous properties of chemical materials. (5)
